



DEPARTMENT OF SOFTWARE ENGINEERING

CLOUD COMPUTING LAB

LAB 01

NAME:

Samreen Zia

REG NO:

FA24B1-SE-067

COURSE CODE:

SE-665

SECTION:

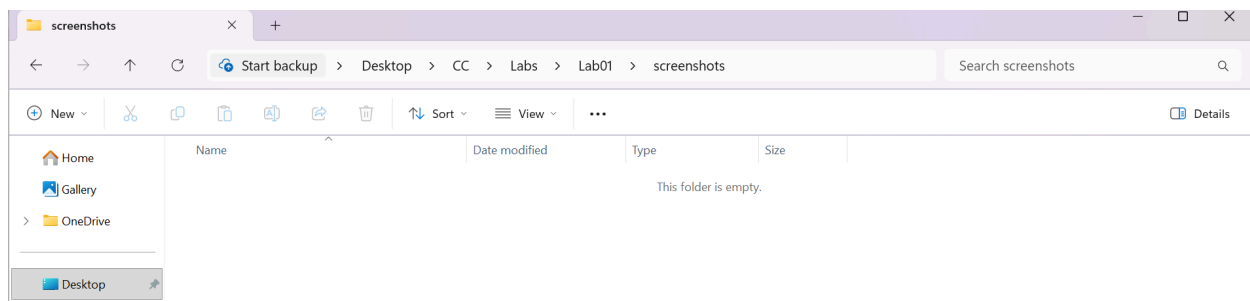
A

GITHUB USER NAME:

samreenzia75

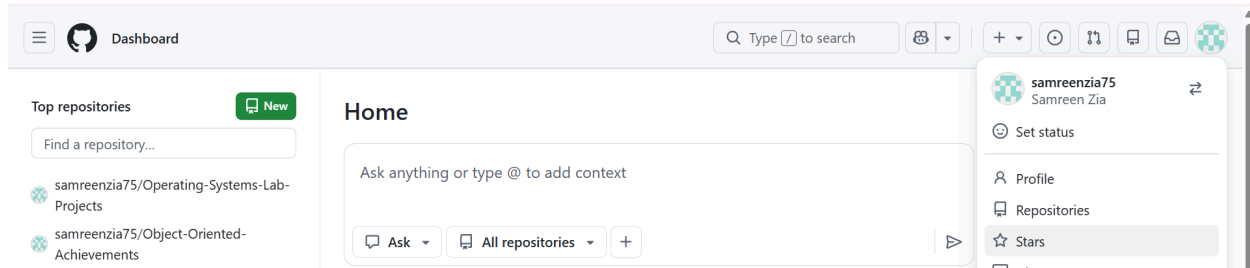
Folder Structure Verification

Demonstrates the creation of the required **Labs/Lab01/screenshots/** directory structure inside the local repository for storing all lab evidence files.

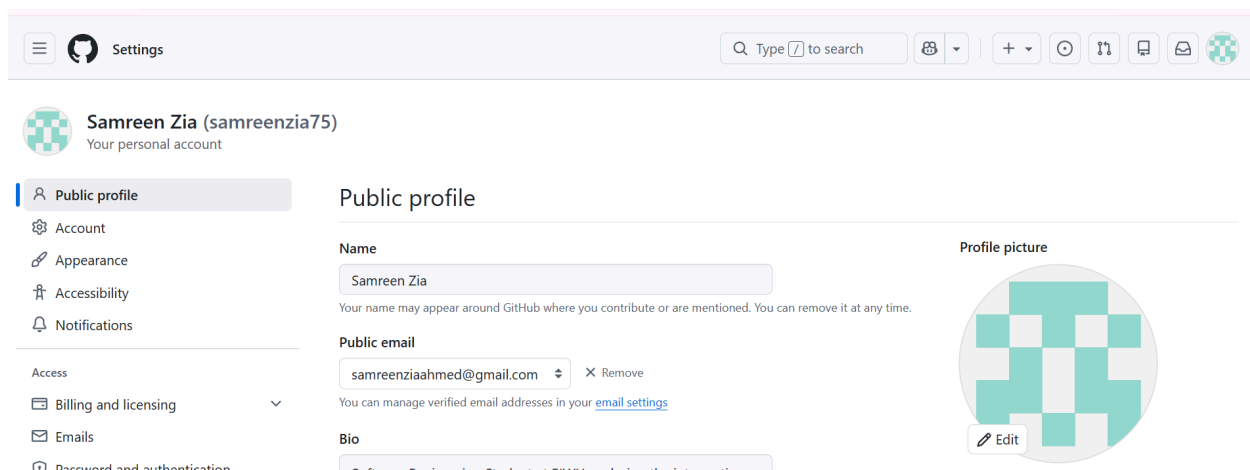


Task 1: Create and Configure Your GitHub Account

Shows the active GitHub account dashboard with the username `samreenzia75` (`github_profile.png`).

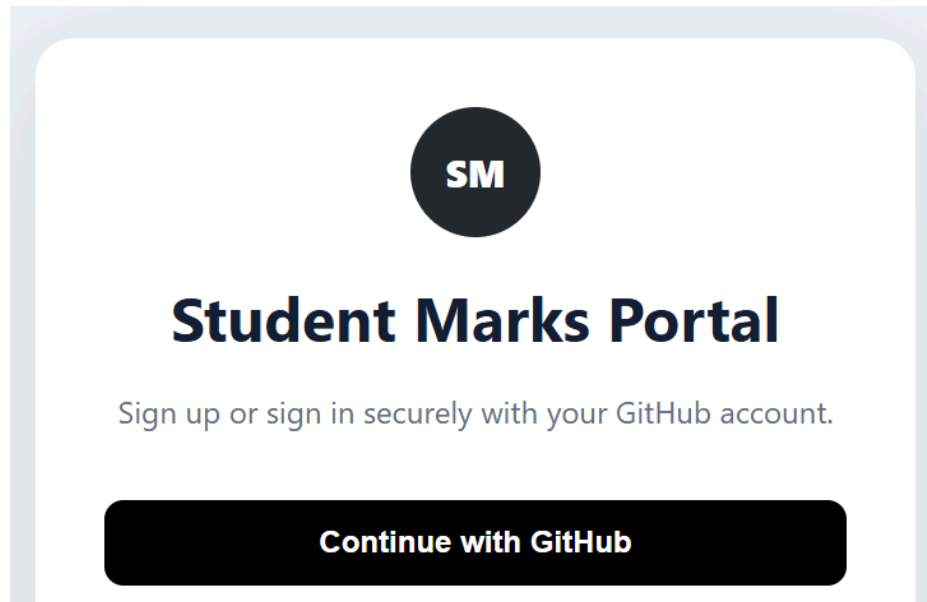


Additionally, displays the GitHub profile settings page confirming the updated full actual name "Samreen Zia" (`github_actual_name.png`).

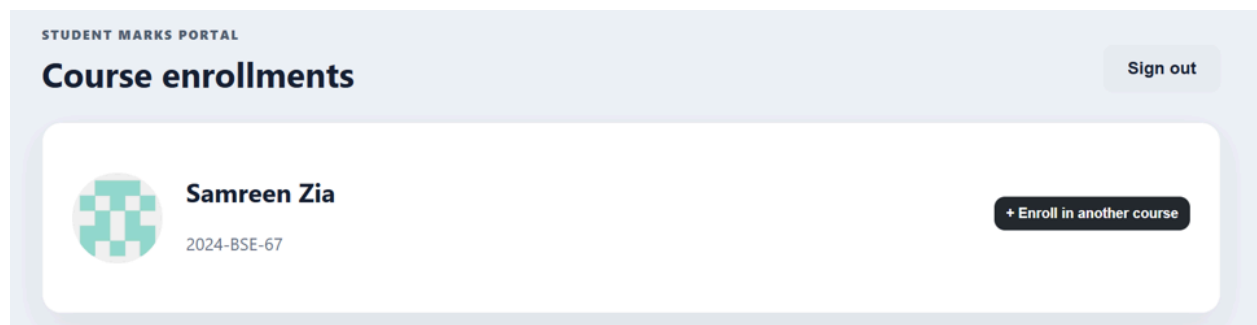


Task 2: Register in the Student Marks Portal

Shows the Student Marks Portal login page with the "Continue with GitHub" sign-in option (`portal_login_page.png`).



Displays the authenticated profile page inside the portal showing student details linked with GitHub ([portal_github_profile.png](#)).

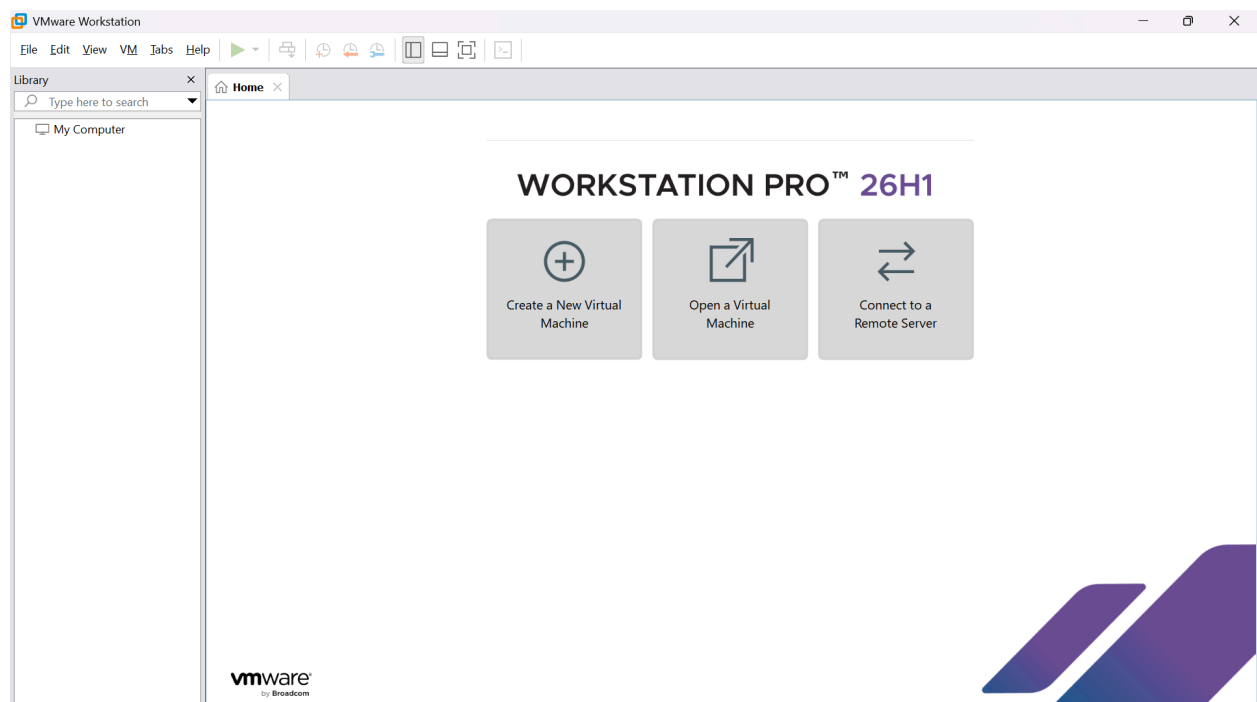


Confirms the submitted course enrollment for Cloud Computing (SE-665) Section A, which is currently pending administrator approval ([portal_enrollment_submitted.png](#)).



Task 3: Download VMware Workstation Pro and Ubuntu Server

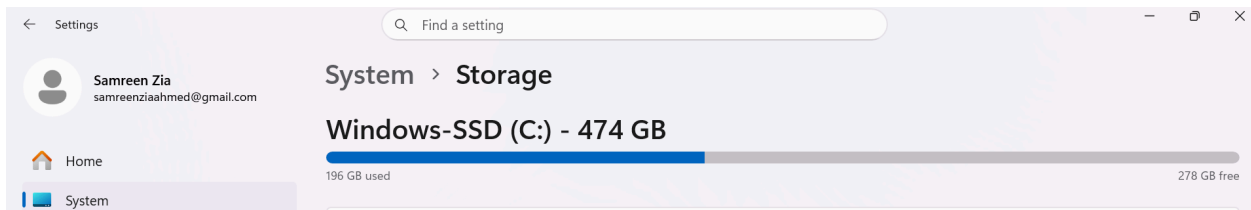
Shows VMware Workstation Pro running on Windows ready for virtual machine creation ([vmware_workstation.png](#)).



Displays the downloaded Ubuntu Server ISO image file stored on the local system ([ubuntu_server_iso.png](#)).

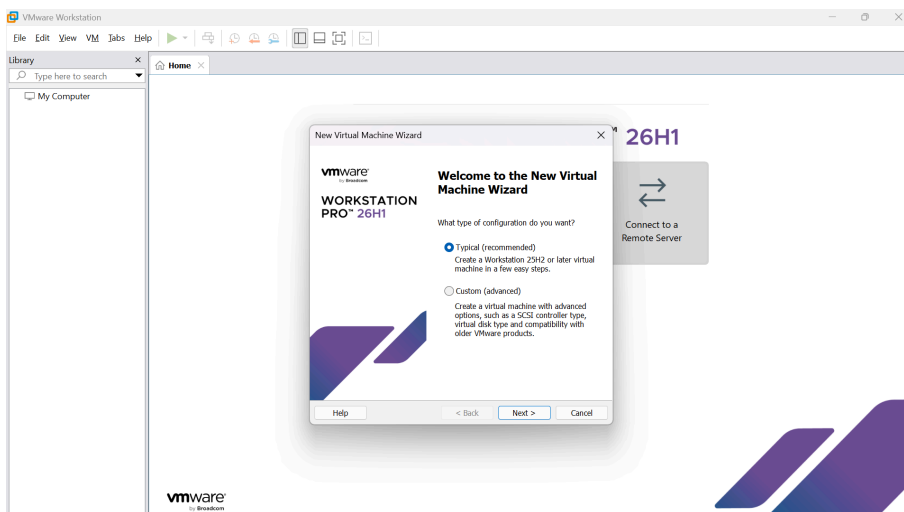
	ubuntu-26.04.1-live-server-amd64 Type: Disc Image File	Date modified: 9/11/2026 11:37 AM Size: 2.72 GB
---	---	--

Verifies that the host system drive has sufficient disk space available (over 20 GB) for virtual machine creation ([available_storage.png](#)).



Task 4: Create the Ubuntu Server Virtual Machine

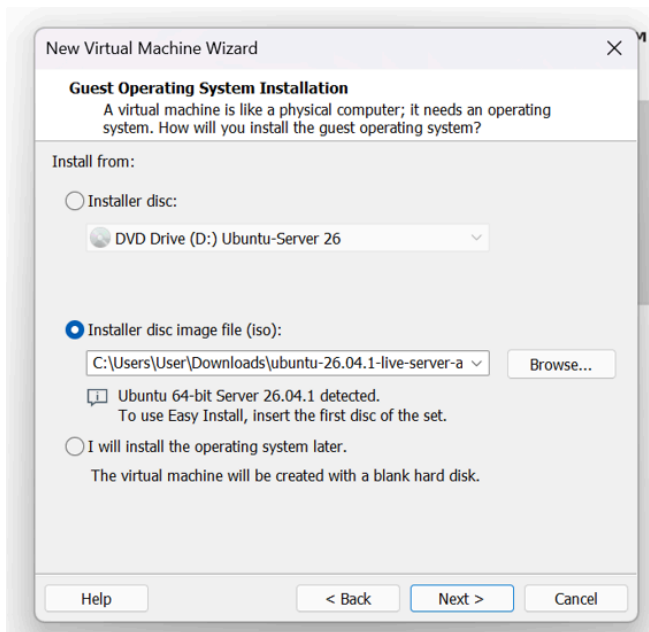
Shows the initial screen of the New Virtual Machine Wizard in VMware Workstation ([vm_creation_wizard.png](#)).



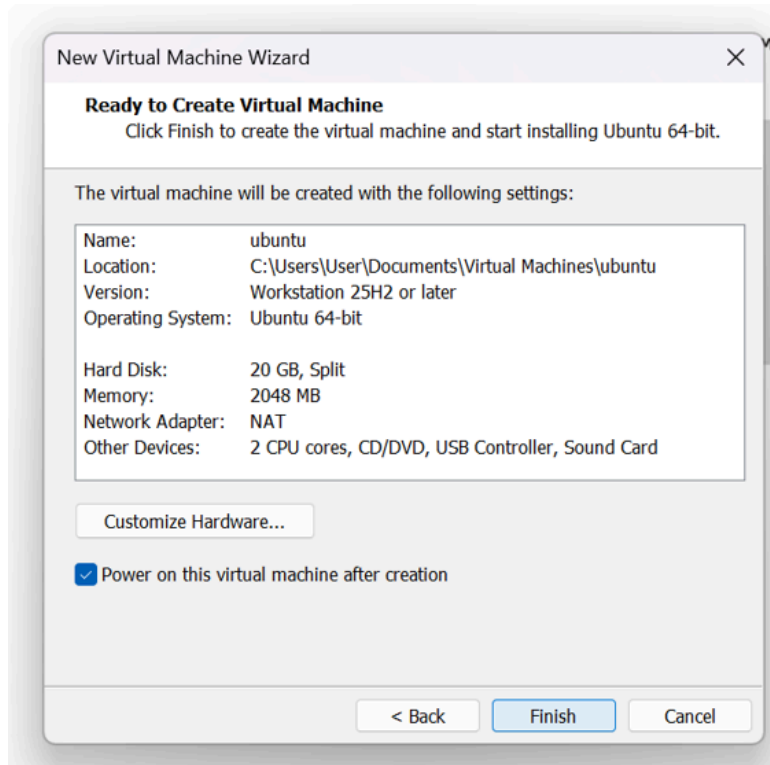
Demonstrates the selection of the Typical (recommended) configuration mode ([vm_typical_configuration.png](#)).



Displays the selected path to the downloaded Ubuntu Server ISO file
([vm_ubuntu_iso_selected.png](#))



Shows the completed VM configuration summary including 20 GB hard disk allocation and memory settings ([vm_configuration_summary.png](#)),

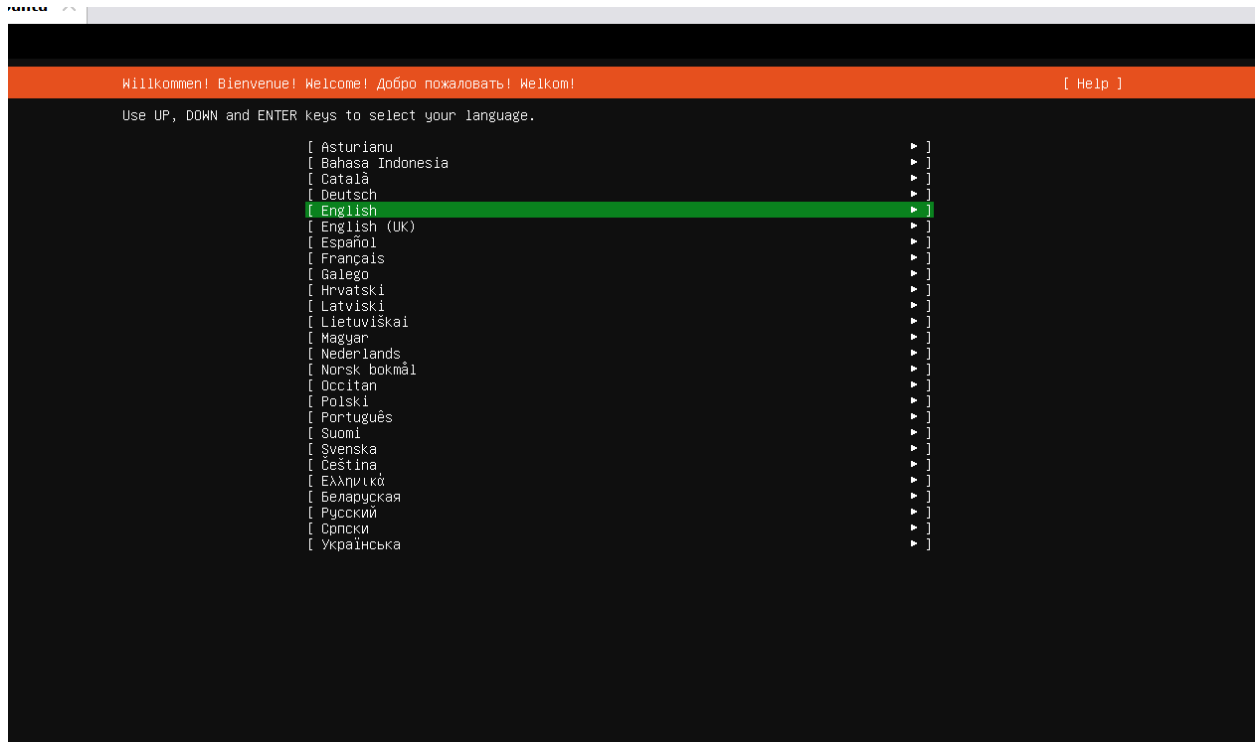


And captures the initial boot sequence of the Ubuntu Server installer inside VMware ([ubuntu_installer_boot.png](#)).

```
[ OK ] Starting rsyslog.service - System Logging Service...
[ OK ] Reached target getty-pre.target - Preparation for Logins.
Starting snapd.seeded.service - Wait until snapd is fully seeded...
Starting snapd.service - Snap Daemon...
Starting ssh.service - OpenBSD Secure Shell server...
Starting sysstat.service - Resets System Activity Logs...
Starting systemd-logind.service - User Login Management...
Starting udisks2.service - Disk Manager...
[ OK ] Finished blk-availability.service - Availability of block devices.
[ OK ] Finished dracut-shutdown.service - Restore /run/initramfs on shutdown.
[ OK ] Finished nvme-autoconnect.service - Connect NVMe-oF subsystems automatically during boot.
[ OK ] Reached target remote-fs-pre.target - Preparation for Remote File Systems.
[ OK ] Reached target remote-fs.target - Remote File Systems.
Starting apport.service - automatic crash report generation...
[ OK ] Started cron.service - Regular background program processing daemon.
Starting multipathd-queueing.service - Enable queuing for multipath maps...
Starting systemd-user-sessions.service - Permit User Sessions...
[ OK ] Started ssh.service - OpenBSD Secure Shell server.
Stopping ssh.service - OpenBSD Secure Shell server...
[ OK ] Stopped ssh.service - OpenBSD Secure Shell server.
Starting ssh.service - OpenBSD Secure Shell server...
[ OK ] Finished multipathd-queueing.service - Enable queuing for multipath maps.
[ OK ] Started ssh.service - OpenBSD Secure Shell server.
[ OK ] Finished cloud-config.service - Cloud-init: Config Stage.
[ OK ] Finished systemd-user-sessions.service - Permit User Sessions.
Starting plymouth-quit-wait.service - Hold until boot process finishes up...
Starting plymouth-quit.service - Terminate Plymouth Boot Screen...
[ OK ] Finished e2scrub_read.service - Remove Stale Online ext4 Metadata Check Snapshots.
[ OK ] Started systemd-logind.service - User Login Management.
[ OK ] Finished sysstat.service - Resets System Activity Logs.
[ OK ] Started unattended-upgrades.service - Unattended Upgrades Shutdown.
[ OK ] Finished plymouth-quit-wait.service - Hold until boot process finishes up.
Starting setvtrgb.service - Set console scheme...
[ OK ] Finished plymouth-quit.service - Terminate Plymouth Boot Screen.
[ OK ] Finished setvtrgb.service - Set console scheme.
[ OK ] Started chrony.service - chrony, an NTP client/server.
[ OK ] Started rsyslog.service - System Logging Service.
[ OK ] Started dbus.service - D-Bus System Message Bus.
[ OK ] Started polkit.service - Authorization Manager.
Starting ModemManager.service - Modem Manager...
[ OK ] Started udisks2.service - Disk Manager.
[ OK ] Started ModemManager.service - Modem Manager.
[ OK ] Finished apport.service - automatic crash report generation.
[ OK ] Started networkd-dispatcher.service - Dispatcher daemon for systemd-networkd.
[ OK ] Started snapd.service - Snap Daemon.
Starting snapd.hold.service - Holds Snappy daemon refresh...
[ OK ] Finished snapd.hold.service - Holds Snappy daemon refresh.
[ OK ] Listening on snap.huclt1.huclt1-daemon.huclt1-socket.socket - Socket huclt1-socket for snap application huclt1.huclt1-daemon.
[ OK ] Started snap.huclt1.hook.configure-bed2b5f8-9c9b-4718-9524-63d0a82f00b4.scope.
```

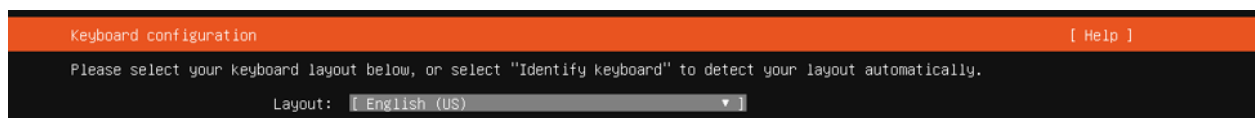
Task 5: Choose the Installation Language

Shows English selected as the primary installation language for Ubuntu Server ([ubuntu_language.png](#)).

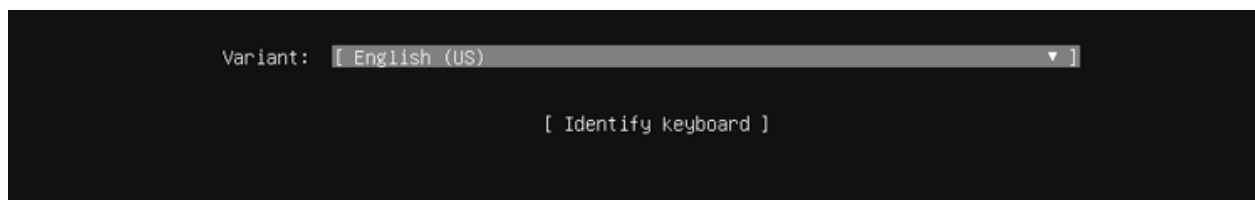


Task 6: Configure the Keyboard Layout

Displays the keyboard configuration step with English (US) selected ([ubuntu_keyboard_layout.png](#)).

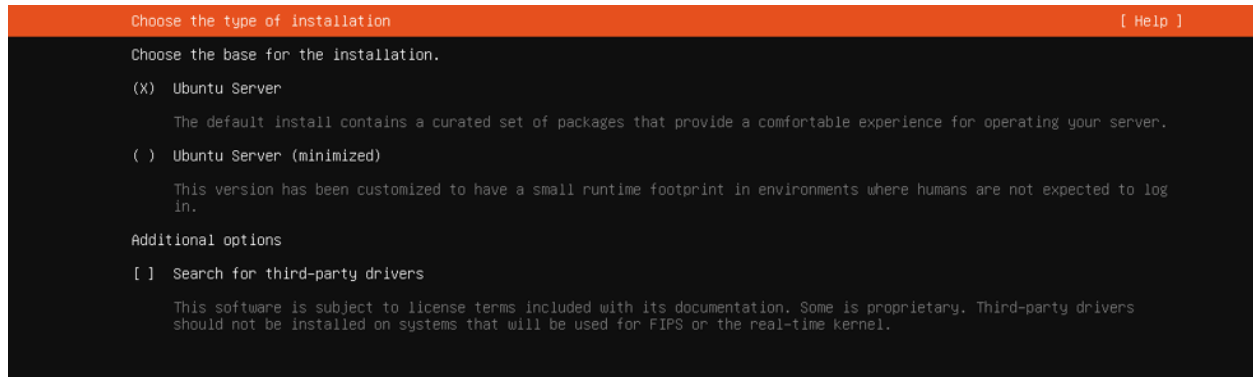


Confirms the selected English (US) keyboard variant ([ubuntu_keyboard_variant.png](#)).



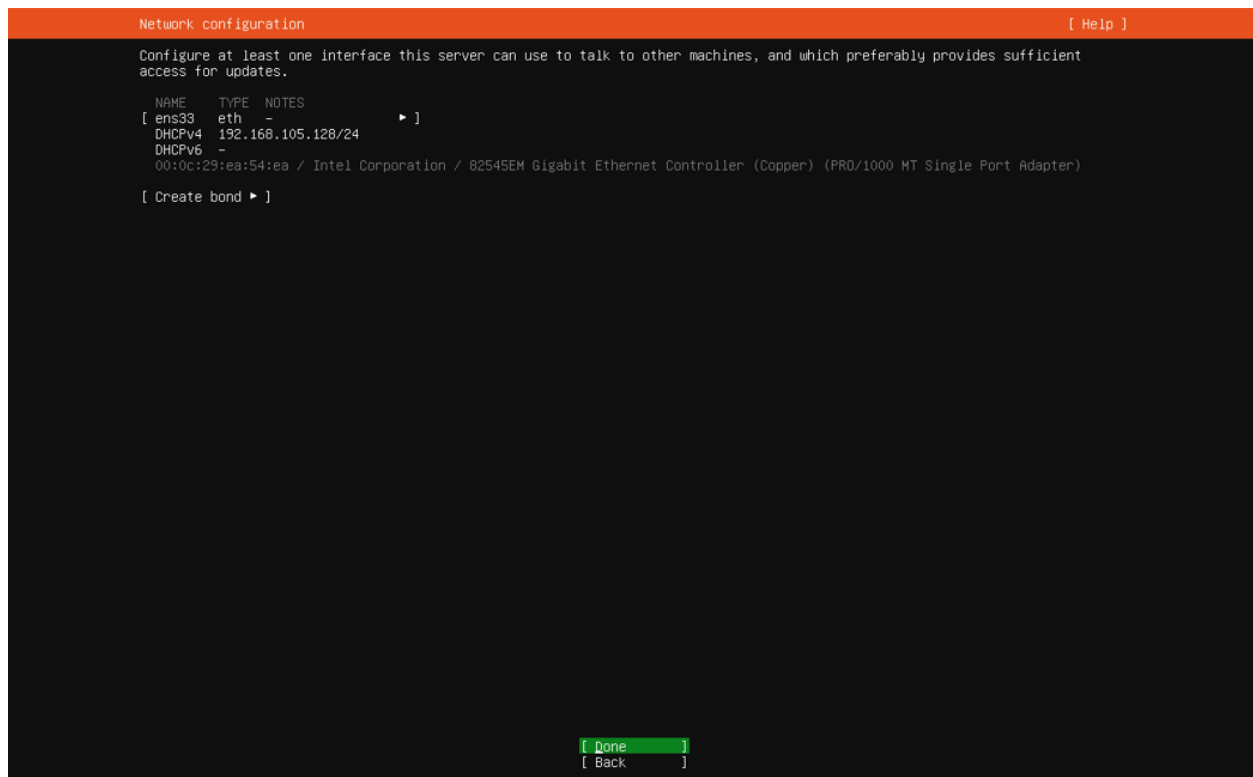
Task 7: Choose the Ubuntu Server Installation Type

Shows the standard "Ubuntu Server" base installation selected instead of the minimized option ([ubuntu_installation_type.png](#)).



Task 8: Configure the Network

Shows the detected network interface ([ens33](#)) configured by default ([ubuntu_network_interface.png](#)).



Demonstrates that the network interface successfully acquired an automatic IP address via DHCP ([ubuntu_installer_network.png](#)).

```
Network configuration [ Help ]

Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient
access for updates.

NAME    TYPE  NOTES
[ ens33  eth  -      ▶ ]
DHCPv4   192.168.105.128/24
DHCPv6   -
00:0c:29:ea:54:ea / Intel Corporation / 82545EM Gigabit Ethernet Controller (Copper) (PRO/1000 MT Single Port Adapter)

[ Create bond ▶ ]
```

Task 9: Configure Storage

Shows the "Use an entire disk" guided storage setup using the allocated virtual hard drive ([ubuntu_guided_storage.png](#)),

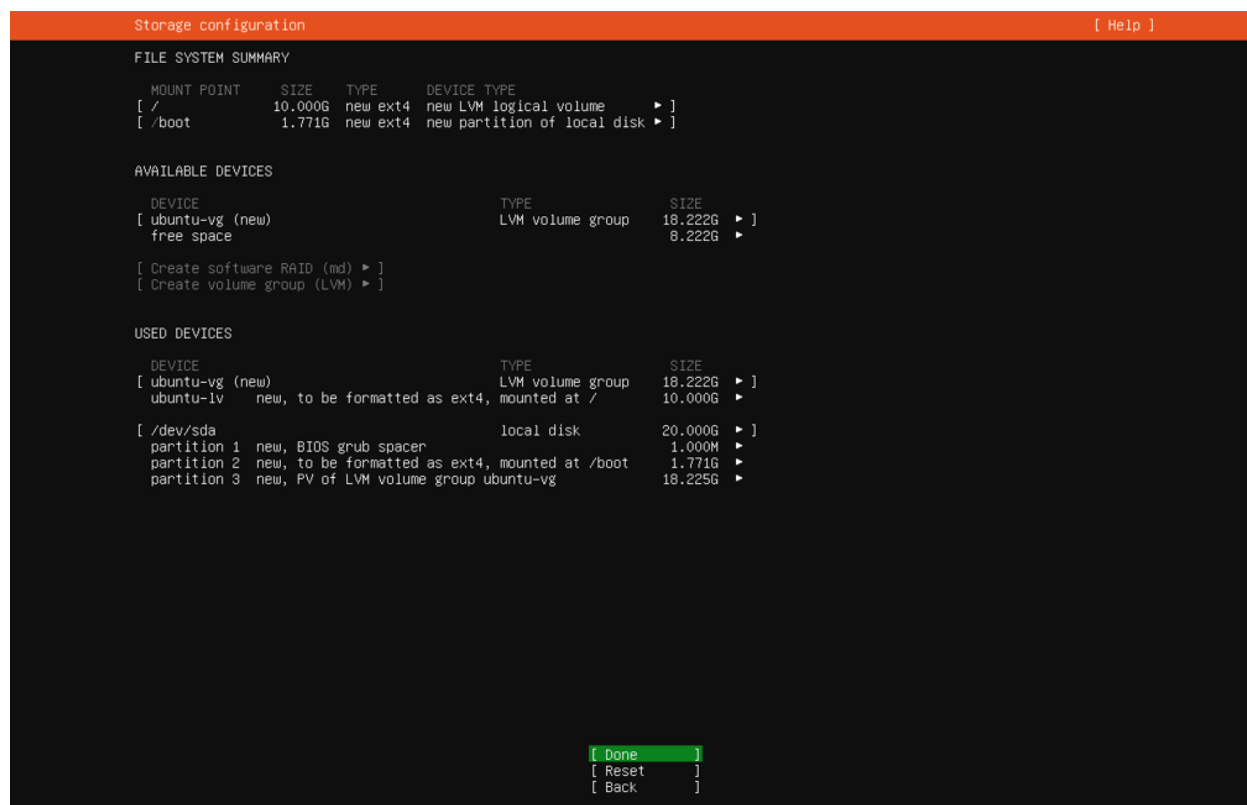
```
Guided storage configuration [ Help ]

Configure a guided storage layout, or create a custom one:

(X) Use an entire disk
    [ /dev/sda local disk 20.000G ▼ ]
    (X) Set up this disk as an LVM group
        [ ] Encrypt the LVM group with LUKS
            Passphrase:
            Confirm passphrase:
            [ ] Also create a recovery key
                The key will be stored as ~/recovery-key.txt in the live system and will be copied to
                /var/log/installer/ in the target system.

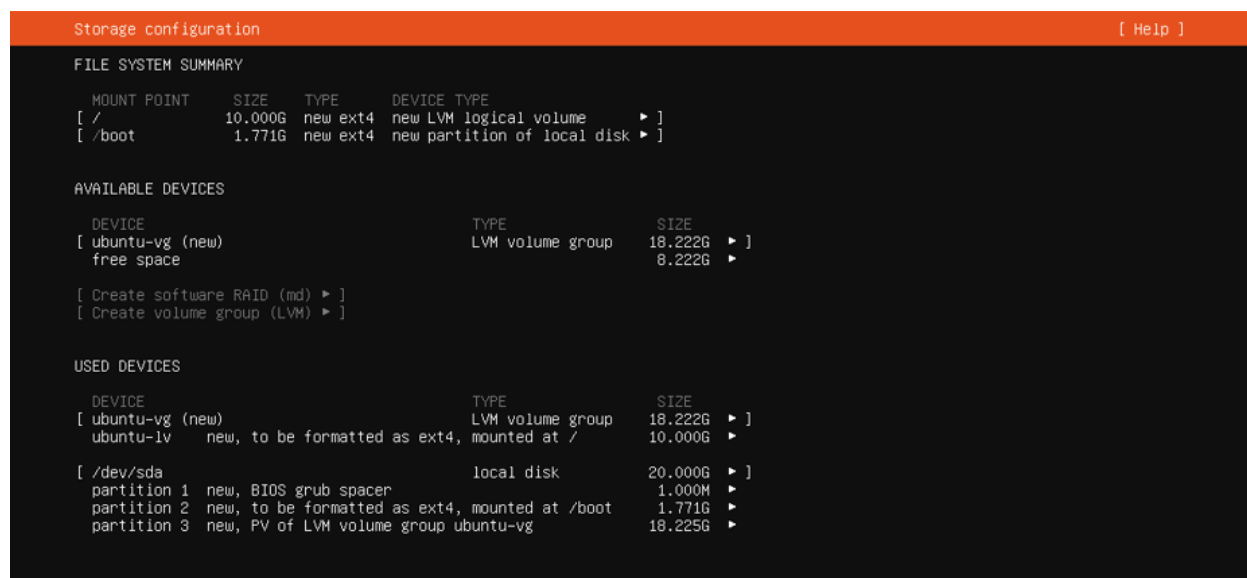
( ) Custom storage layout
```

Displays the file system summary and LVM volume group layout proposed by the installer ([ubuntu_storage_configuration.png](#)).

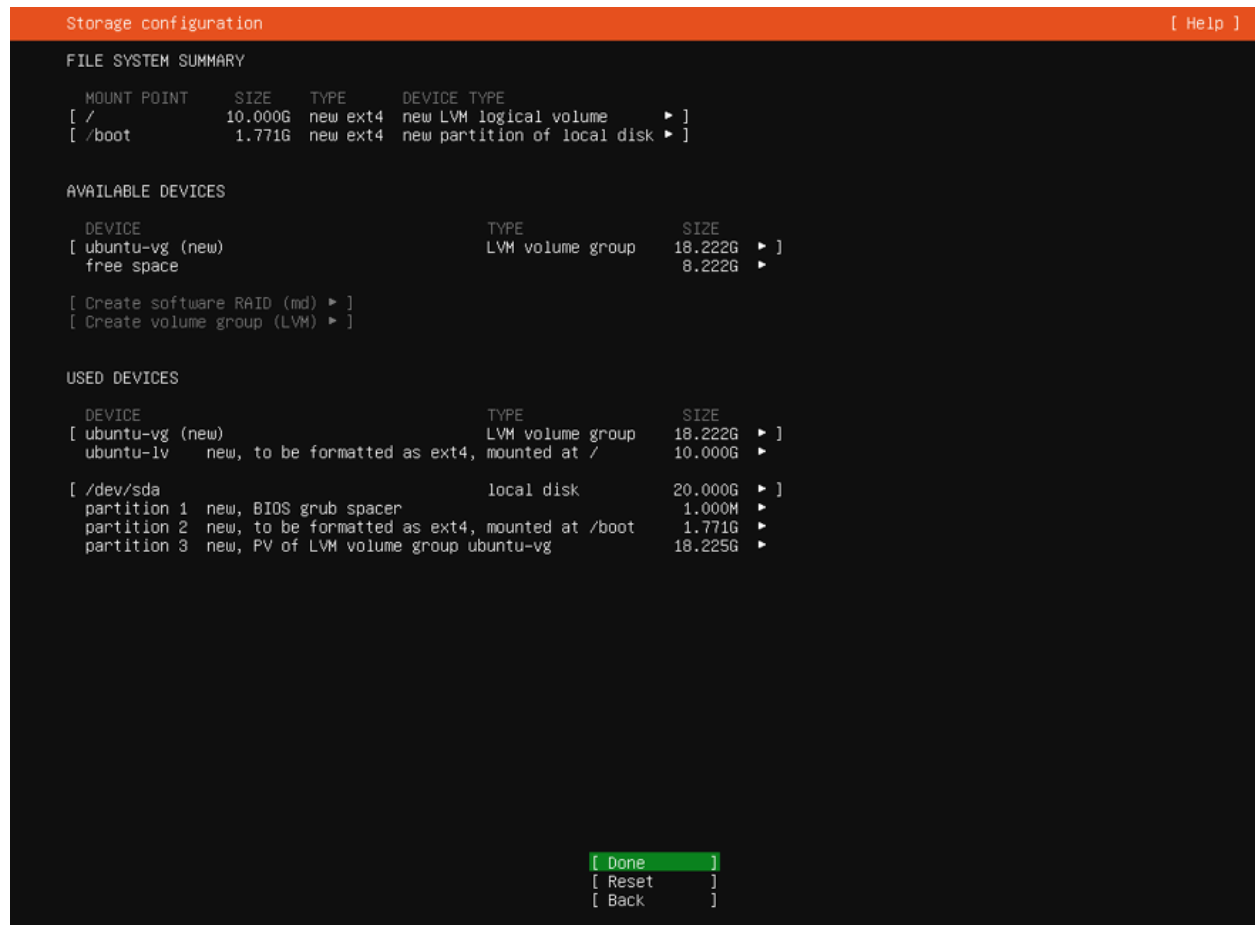


Task 10: Select the Installation Disk

Highlights the selection of the 20 GB VMware local disk `/dev/sda` (`ubuntu_selected_disk.png`).

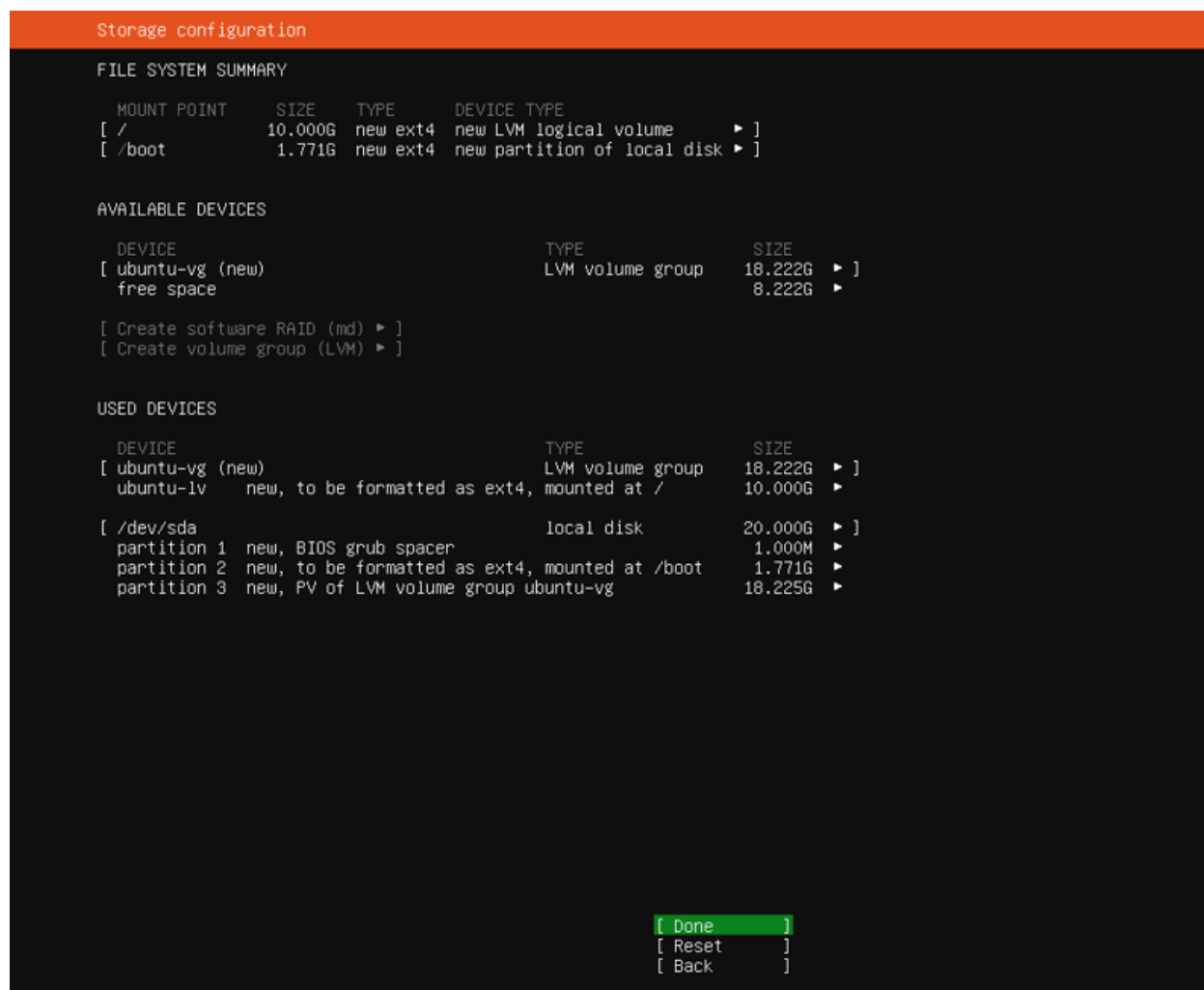


Confirms the exact size and partition layout details of the selected virtual disk ([ubuntu_disk_details.png](#)).



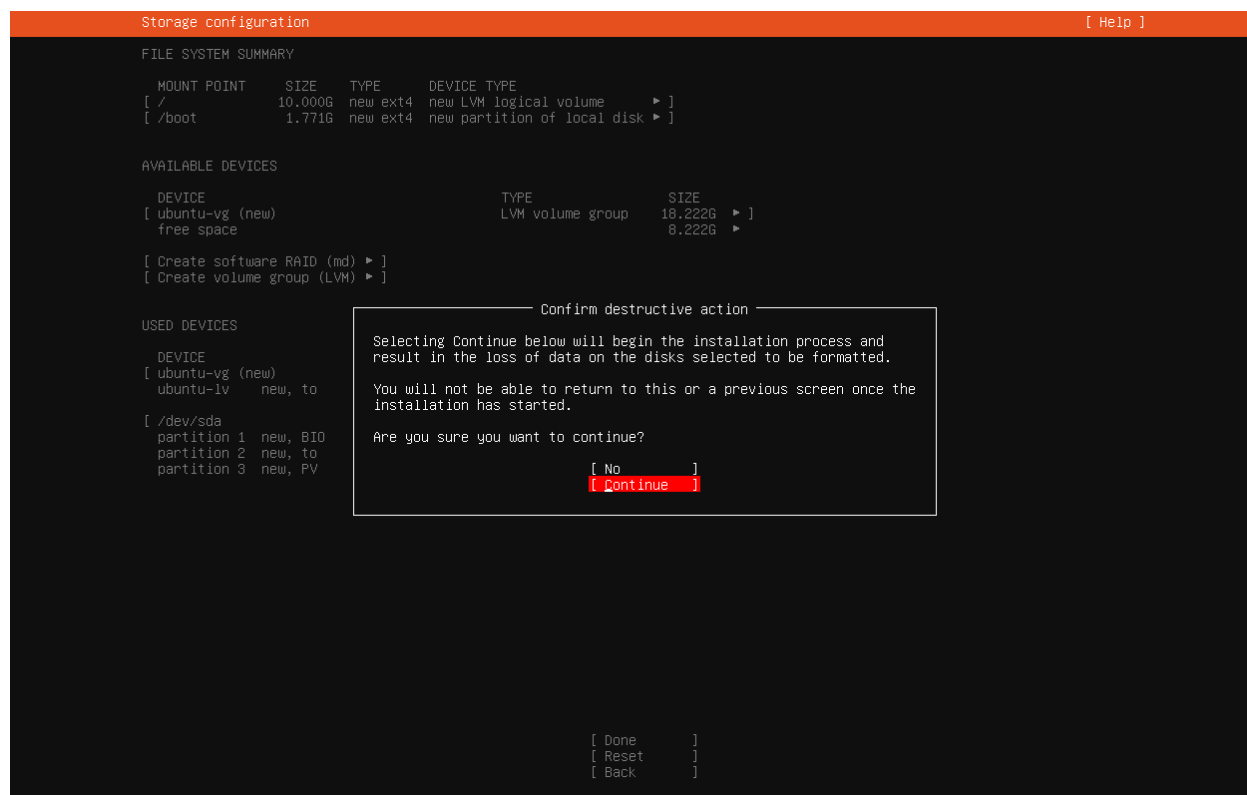
Task 11: Review the Partitions

Displays the full partition structure including **/boot** and LVM root partitions before applying changes ([ubuntu_partition_layout.png](#)).

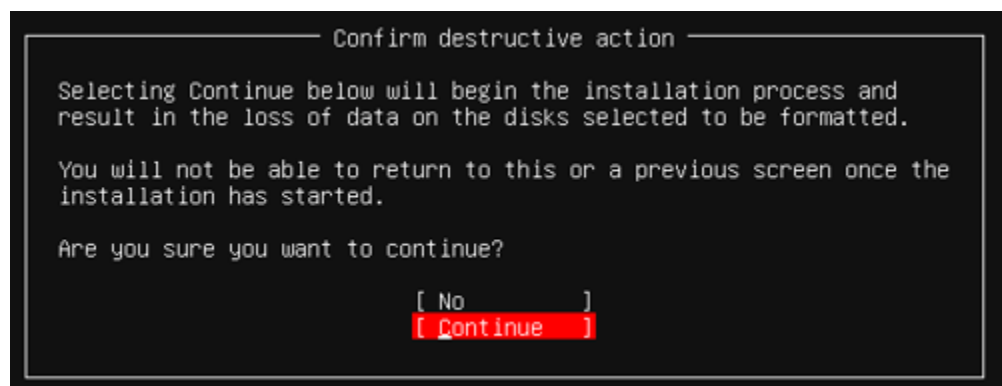


Task 12: Confirm the Storage Changes

Shows the destructive action confirmation prompt warning about formatting the virtual disk ([ubuntu_storage_warning.png](#))

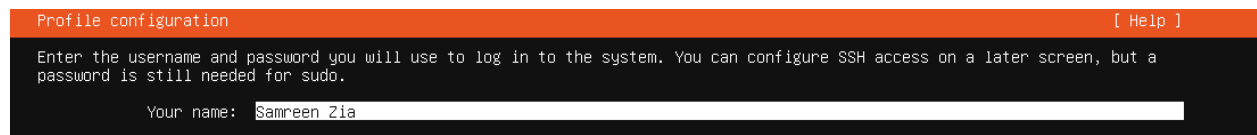


And demonstrates selecting "Continue" to proceed with disk partitioning (ubuntu_storage_confirmed.png).



Task 13: Configure the Ubuntu Profile

Shows the "Your name" field populated with "Samreen Zia" (ubuntu_actual_name.png).



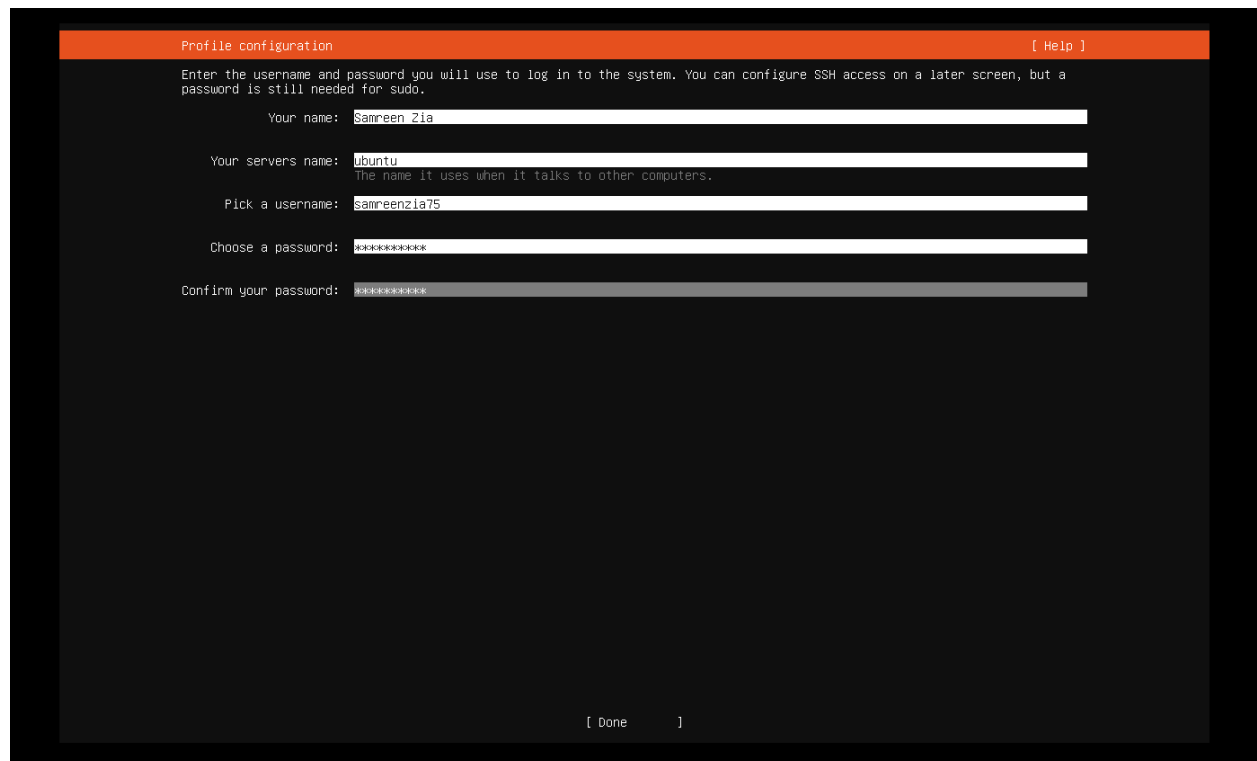
The "Your server's name" field configured as `ubuntu` (`ubuntu_server_name.png`).



Demonstrates the username field set to match the GitHub username `samreenzia75` (`ubuntu_username.png`).

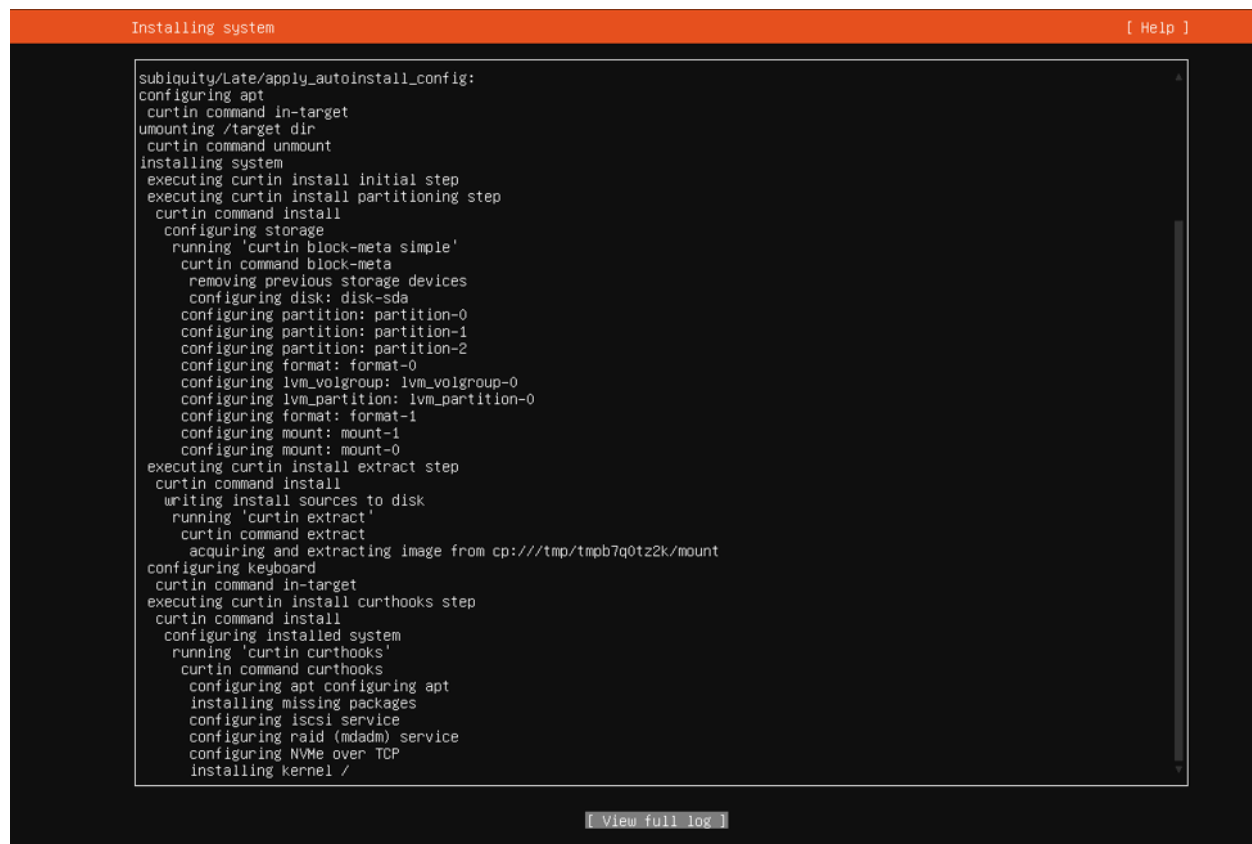


Displays the fully completed profile setup page including confirmed password fields (`ubuntu_profile_setup.png`).



Task 14: Complete the Software Installation

Captures the progress screen while Ubuntu Server extracts and installs base packages and system files (`ubuntu_installation_progress.png`).



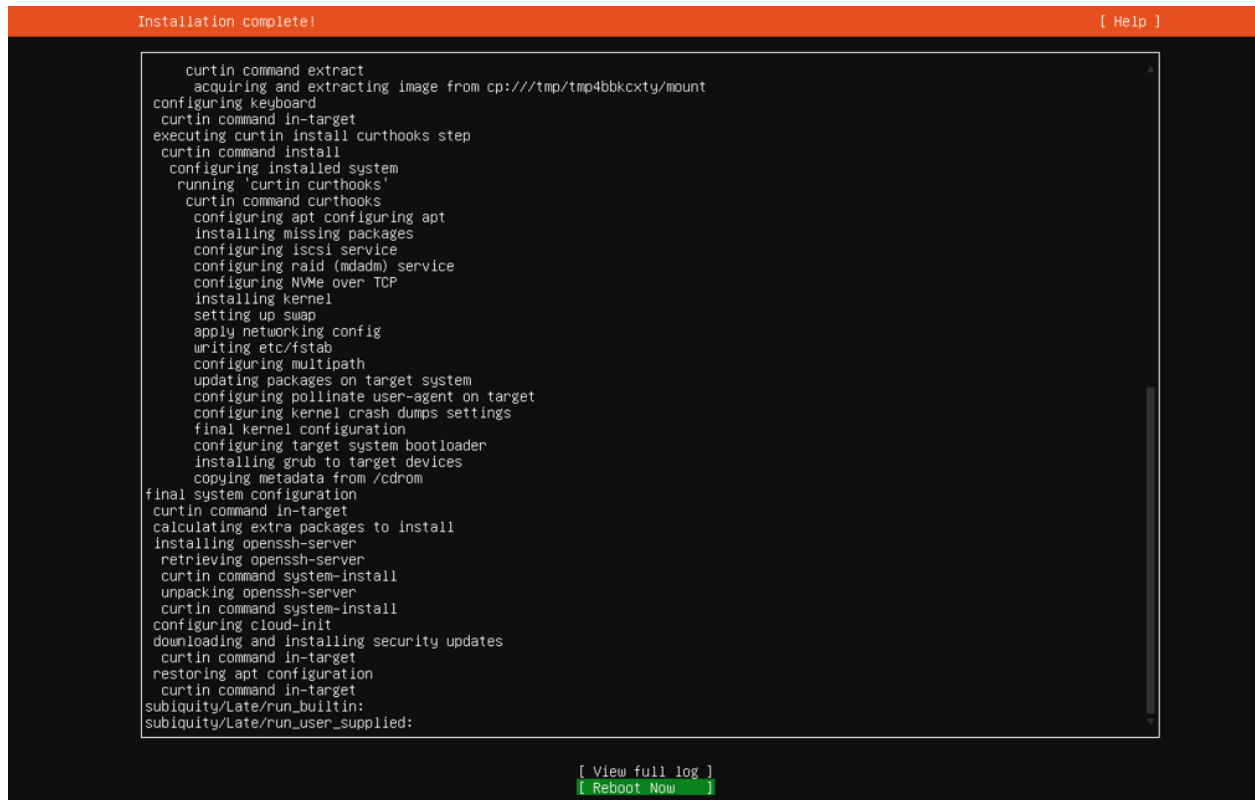
The screenshot shows a terminal window titled "Installing system" with a "[Help]" button in the top right corner. The terminal displays a series of commands and their outputs, indicating the progress of the Ubuntu installation. The commands include configuring apt, mounting and unmounting the target directory, installing the system, configuring storage (disk-sda, partitions, format, lvm), executing the install initial and partitioning steps, configuring the installed system (keyboard, curthooks, apt, missing packages, iSCSI service, RAID service, NVMe over TCP), and installing the kernel. A "[View full log]" button is located at the bottom center of the terminal window.

```
subiquity/late/apply_autoinstall_config:
configuring apt
  curtin command in-target
umounting /target dir
  curtin command unmount
installing system
  executing curtin install initial step
  executing curtin install partitioning step
  curtin command install
    configuring storage
      running 'curtin block-meta simple'
      curtin command block-meta
        removing previous storage devices
        configuring disk: disk-sda
        configuring partition: partition-0
        configuring partition: partition-1
        configuring partition: partition-2
        configuring format: format-0
        configuring lvm_volgroup: lvm_volgroup-0
        configuring lvm_partition: lvm_partition-0
        configuring format: format-1
        configuring mount: mount-1
        configuring mount: mount-0
    executing curtin install extract step
    curtin command install
      writing install sources to disk
      running 'curtin extract'
      curtin command extract
        acquiring and extracting image from cp:///tmp/tmpb7q0tz2k/mount
    configuring keyboard
    curtin command in-target
  executing curtin install curthooks step
  curtin command install
    configuring installed system
    running 'curtin curthooks'
    curtin command curthooks
      configuring apt configuring apt
      installing missing packages
      configuring iSCSI service
      configuring RAID (mdadm) service
      configuring NVMe over TCP
    installing kernel /
```

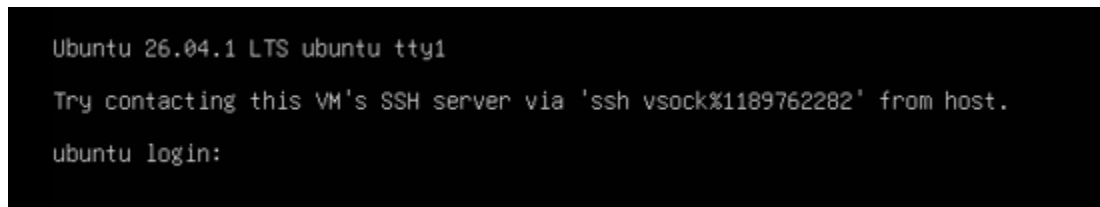
[View full log]

Task 15: Reboot and Verify the Ubuntu Identity

Displays the installer status indicating that installation is complete and ready to reboot ([ubuntu_installation_complete.png](#)).



Shows the login prompt of the newly booted Ubuntu Server instance (ubuntu_login_screen.png).



Demonstrates successful authentication into the server using the configured credentials (ubuntu_terminal_login.png).

```
Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%2727716573' from host.

ubuntu login: samreenzia75
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Aug 26 04:51:42 UTC 2026

System load: 1.07      Processes:            26
Usage of /home: unknown  Users logged in:      0
Memory usage: 8%      IPv4 address for eth0: 10.10.10.2
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

samreenzia75@ubuntu:~$
```

Confirms that the active terminal shell prompt strictly follows the required `samreenzia75@ubuntu:~$` format (`ubuntu_identity_verified.png`).

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

samreenzia75@ubuntu:~$
```

Task 16: Find the Ubuntu Server IP Address

Displays the execution of the `ip addr` command in the Ubuntu terminal (`ubuntu_ip_addr_command.png`).

```
samreenzia75@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:95:a6:dd brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2995a6dd
    inet 192.168.121.129/24 metric 100 brd 192.168.121.255 scope global dynamic ens33
        valid_lft 1705sec preferred_lft 1705sec
    inet6 fe80::20c:29ff:fe95:a6dd/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
samreenzia75@ubuntu:~$ _
```

Highlights the assigned IPv4 address (192.168.121.129) under interface ens33 while maintaining prompt visibility (ubuntu_ip_address.png).

```
samreenzia75@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:95:a6:dd brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2995a6dd
    inet 192.168.121.129/24 metric 100 brd 192.168.121.255 scope global dynamic ens33
        valid_lft 1705sec preferred_lft 1705sec
    inet6 fe80::20c:29ff:fe95:a6dd/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Task 17: Connect to Ubuntu Server from Windows

Shows the SSH command `ssh samreenzia75@192.168.121.129` executed from Windows Command Prompt (windows_ssh_command.png).

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh samreenzia75@192.168.121.129
The authenticity of host '192.168.121.129 (192.168.121.129)' can't be established.
ED25519 key fingerprint is SHA256:AgAZxGxa572FLVSC3/kag1WT8mmAH5vt77ic3tYvQhM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

Demonstrates accepting the host ED25519 key fingerprint by entering `yes` (windows_ssh_fingerprint.png).

```
samreenzia75@ubuntu: ~  
Microsoft Windows [Version 10.0.26200.9457]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\User>ssh samreenzia75@192.168.121.129  
The authenticity of host '192.168.121.129 (192.168.121.129)' can't be established.  
ED25519 key fingerprint is SHA256:AgAZxGxa572FLVSC3/kag1WT8mmAH5vt77ic3tYvQhM.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.121.129' (ED25519) to the list of known hosts.
```

Confirms successful remote password login to Ubuntu Server over SSH
([windows_ssh_login.png](#)).

```
samreenzia75@ubuntu: ~  
Microsoft Windows [Version 10.0.26200.9457]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\User>ssh samreenzia75@192.168.121.129  
The authenticity of host '192.168.121.129 (192.168.121.129)' can't be established.  
ED25519 key fingerprint is SHA256:AgAZxGxa572FLVSC3/kag1WT8mmAH5vt77ic3tYvQhM.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.121.129' (ED25519) to the list of known hosts.  
samreenzia75@192.168.121.129's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Fri Sep 18 08:07:09 AM UTC 2026  
  
System load:  0.02      Processes:            220  
Usage of /:   46.0% of 9.75GB  Users logged in:     0  
Memory usage: 22%      IPv4 address for ens33: 192.168.121.129  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

Verifies that the remote session shell displays the correct prompt `samreenzia75@ubuntu:~$`
over SSH ([windows_ssh_identity.png](#)).

```
samreenzia75@ubuntu: ~
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh samreenzia75@192.168.121.129
The authenticity of host '192.168.121.129 (192.168.121.129)' can't be established.
ED25519 key fingerprint is SHA256:AgAZxGxa572FLVSC3/kag1WT8mmAH5vt77ic3tYvQhM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.121.129' (ED25519) to the list of known hosts.
samreenzia75@192.168.121.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 18 08:07:09 AM UTC 2026

System load:  0.02          Processes:            220
Usage of /:   46.0% of 9.75GB Users logged in:        0
Memory usage: 22%          IPv4 address for ens33: 192.168.121.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

samreenzia75@ubuntu:~$ |
```

Task 18: Prepare Lab1 Solution PDF

Shows the title page containing lab details, actual name, registration number, course code, section, and GitHub username ([solution_title_page.png](#)).

DEPARTMENT OF SOFTWARE ENGINEERING

CLOUD COMPUTING LAB

LAB 01

NAME:

Samreen Zia

REG NO:

FA24B1-SE-067

COURSE CODE:

SE-665

SECTION:

A

GITHUB USER NAME:

samreenzia75

Confirms that the final compiled document was exported as **Lab1_Solution.pdf** and placed in **Labs/Lab01/ (lab1_solution_pdf.png)**.

